

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
```

```
df = pd.read_csv("Mall_Customers.csv")
```

```
# Display first few rows
print("First 5 rows:")
print(df.head())
```

```
# Display dataset information
print("\nDataset Information:")
print(df.info())
```

```
First 5 rows:
   CustomerID  Genre  Age  Annual Income (k$)  Spending Score (1-100)
0           1   Male   19                15                39
1           2   Male   21                15                81
2           3  Female  20                16                 6
3           4  Female  23                16                77
4           5  Female  31                17                40
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column              Non-Null Count  Dtype
---  -
 0   CustomerID          200 non-null   int64
 1   Genre               200 non-null   object
 2   Age                 200 non-null   int64
 3   Annual Income (k$)  200 non-null   int64
 4   Spending Score (1-100) 200 non-null   int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
None
```

```
# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Remove duplicate records
df = df.drop_duplicates()

# Select relevant features
X = df[['Age', 'Annual Income (k$)', 'Spending Score (1-100)']]

print("\nSelected Features:")
print(X.head())
```

```
Missing Values:
CustomerID      0
Genre           0
Age             0
Annual Income (k$) 0
Spending Score (1-100) 0
dtype: int64

Selected Features:
   Age  Annual Income (k$)  Spending Score (1-100)
0   19                 15                39
1   21                 15                81
2   20                 16                 6
3   23                 16                77
4   31                 17                40
```

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
```

```
wcss = []

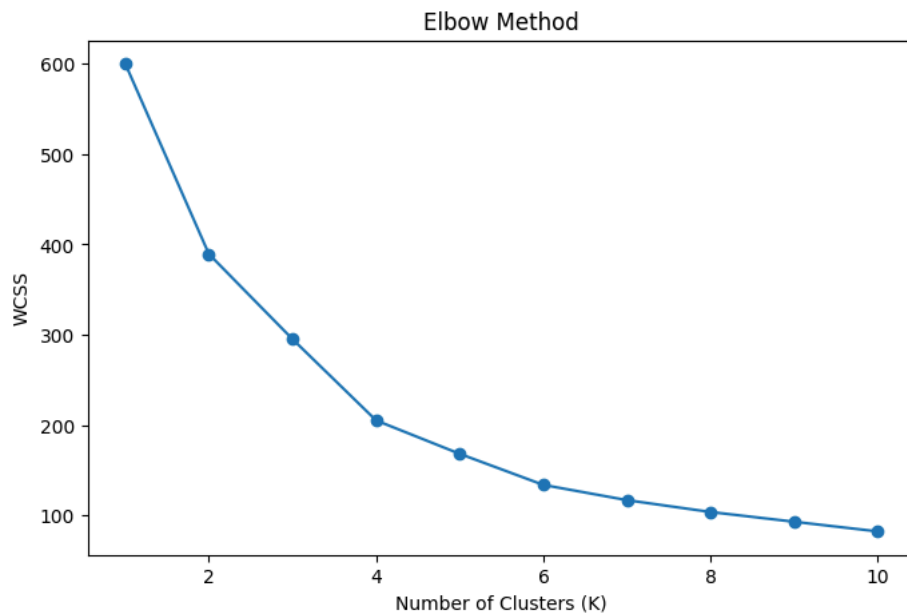
for k in range(1, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
```

```

kmeans.fit(X_scaled)
wcss.append(kmeans.inertia_)

# Plot Elbow Curve
plt.figure(figsize=(8, 5))
plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.show()

```



```

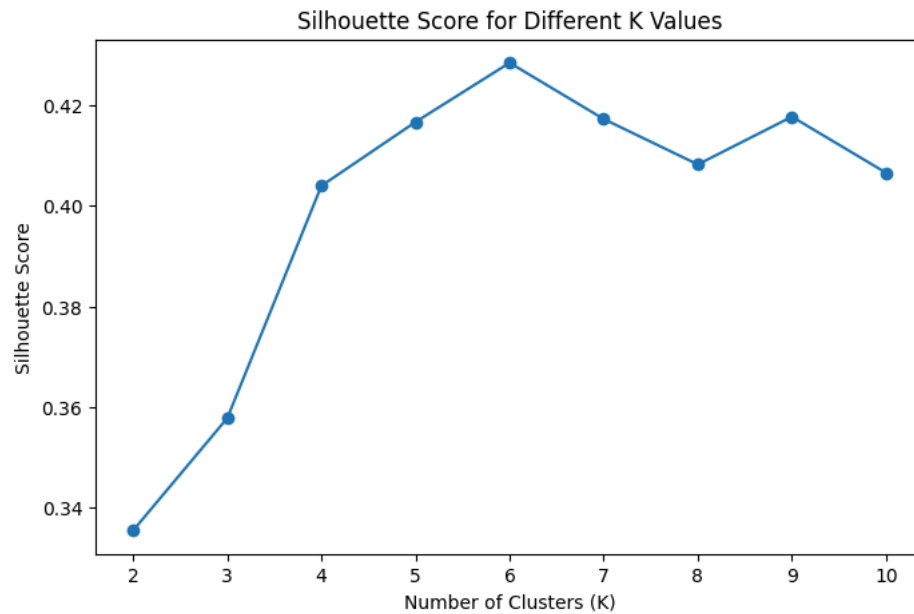
silhouette_scores = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    labels = kmeans.fit_predict(X_scaled)
    score = silhouette_score(X_scaled, labels)
    silhouette_scores.append(score)

# Plot Silhouette Scores
plt.figure(figsize=(8, 5))
plt.plot(range(2, 11), silhouette_scores, marker='o')
plt.title("Silhouette Score for Different K Values")
plt.xlabel("Number of Clusters (K)")
plt.ylabel("Silhouette Score")
plt.show()

# Print scores
for k, score in zip(range(2, 11), silhouette_scores):
    print(f"K = {k}, Silhouette Score = {score:.4f}")

```



```

K = 2, Silhouette Score = 0.3355
K = 3, Silhouette Score = 0.3578
K = 4, Silhouette Score = 0.4040
K = 5, Silhouette Score = 0.4166
K = 6, Silhouette Score = 0.4284
K = 7, Silhouette Score = 0.4172
K = 8, Silhouette Score = 0.4082
K = 9, Silhouette Score = 0.4177
K = 10, Silhouette Score = 0.4066

```

```

# For the Mall Customer dataset, K=5 is commonly used.
k = 5

kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)

df['Cluster'] = kmeans.fit_predict(X_scaled)

```

```

print("\nClustered Data:")
print(df.head())

# 9. Cluster Centroids
centroids = scaler.inverse_transform(kmeans.cluster_centers_)

centroid_df = pd.DataFrame(
    centroids,
    columns=['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
)

print("\nCluster Centroids:")
print(centroid_df)

```

```

Clustered Data:
   CustomerID  Genre  Age  Annual Income (k$)  Spending Score (1-100) \
0           1   Male   19              15              39
1           2   Male   21              15              81
2           3  Female   20              16               6
3           4  Female   23              16              77
4           5  Female   31              17              40

```

```

Cluster
0      1
1      1
2      0
3      1
4      1

```

```

Cluster Centroids:
   Age  Annual Income (k$)  Spending Score (1-100)
0  46.250000      26.750000      18.350000
1  25.185185      41.092593      62.240741
2  32.875000      86.100000      81.525000
3  39.871795      86.102564      19.358974
4  55.638298      54.382979      48.851064

```

```

plt.figure(figsize=(10, 7))

sns.scatterplot(

```

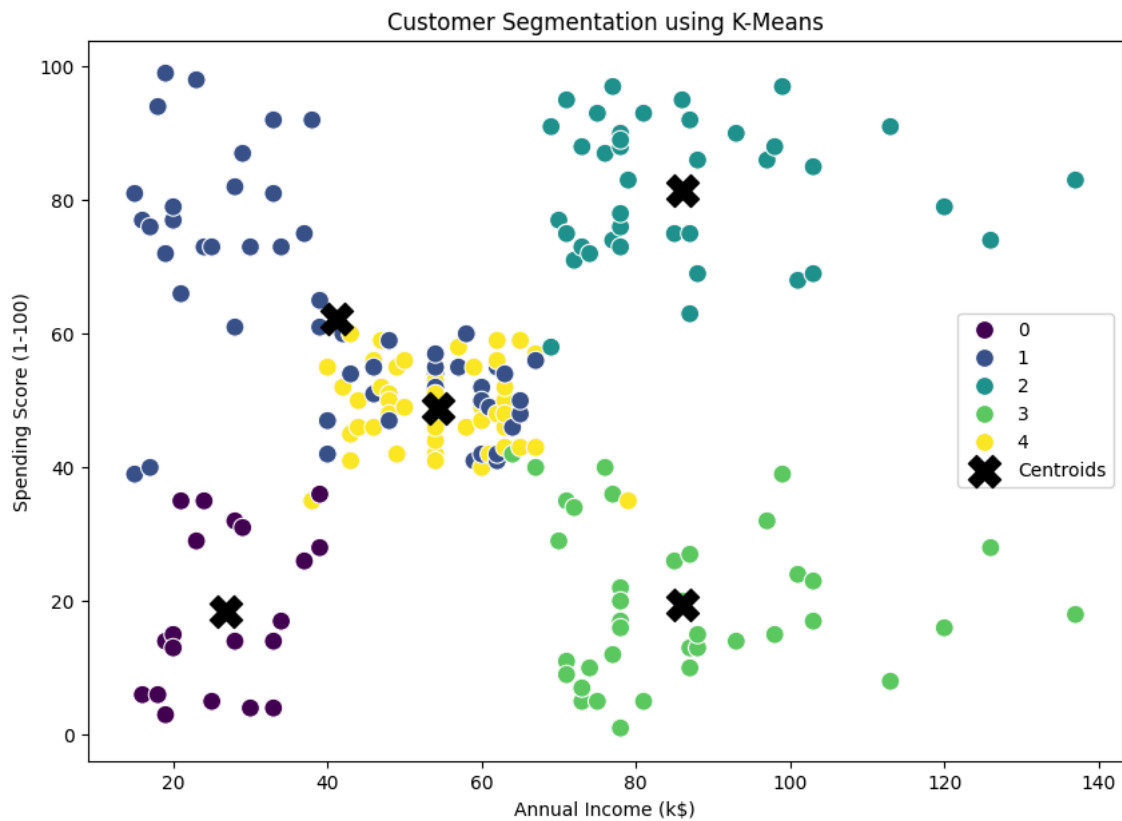
```

data=df,
x='Annual Income (k$)',
y='Spending Score (1-100)',
hue='Cluster',
palette='viridis',
s=100
)

# Plot centroids
plt.scatter(
    centroids[:, 1],
    centroids[:, 2],
    marker='X',
    s=300,
    c='black',
    label='Centroids'
)

plt.title("Customer Segmentation using K-Means")
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.legend()
plt.show()

```



```

print("\nCustomer Segment Analysis:")

cluster_summary = df.groupby('Cluster')[
    ['Age', 'Annual Income (k$)', 'Spending Score (1-100)']
].mean()

print(cluster_summary)

# Number of customers in each cluster
print("\nNumber of Customers in Each Cluster:")
print(df['Cluster'].value_counts().sort_index())

```

```

Customer Segment Analysis:
   Age  Annual Income (k$)  Spending Score (1-100)
Cluster
0      46.250000          26.750000             18.350000
1      25.185185          41.092593             62.240741
2      32.875000          86.100000             81.525000
3      39.871795          86.102564             19.358974
4      55.638298          54.382979             48.851064

Number of Customers in Each Cluster:
Cluster
0      20

```

```
1    54
2    40
3    39
4    47
Name: count, dtype: int64
```

```
df.to_csv("Customer_Segmentation_Result.csv", index=False)
```